Vogel · Fetkovich · Hagedorn-Brown · Brent-method operating point

Production engineering — nodal analysis & well deliverability

A node-based well-performance engine: intersect an inflow (IPR) curve with a vertical-lift (VLP) pressure traverse to find the operating point, then bound it with a confidence band derived from the underlying test-quality score. The correlations are textbook and each is unit-tested against its published equation; the distinctive layer is physics-based data reconciliation — GIGO divergence and nonlinearity flags that catch a bad test record before it corrupts the nodal model. Distinct from the rod-pump dynacard diagnostics under Artificial lift.

±5 / ±15 / ±30%
confidence bands (G/A/R)

≥20%
GIGO divergence flag

WHAT YOU GET

- Operating point by solving IPR(q) = VLP(q) with Brent's method, bracketed from zero to absolute open flow
- Vogel / Fetkovich / Klins-Clark inflow and Hagedorn-Brown / Beggs-Brill outflow correlations, each unit-tested against its published equation
- Green / Amber / Red confidence band keyed to the test-quality score
- ESP sizing from total dynamic head (net lift + Hazen-Williams tubing friction + discharge head) with stage-count, motor and operating-range pass/fail
- Test-to-model reconciliation: QC score, nonlinearity detection (transient / slug / gas-lift instability / high watercut), GIGO divergence diagnosis and recommendations